



Mortgage Portfolio Analytics

Dashboard - Complete Overview

What This Dashboard Represents

This is a comprehensive mortgage portfolio management and risk analytics platform designed to replicate the functionality of enterprise-level tools like Microsoft Fabric, Power BI, and Azure Data Factory. It represents a **\$658+ million** mortgage portfolio with detailed analytics for financial institutions, mortgage lenders, and investment firms.



Dataset Overview

Portfolio Composition:

- ❖ 750 Mortgage Loans across Vancouver and Metro Vancouver regions
- ❖ Total Portfolio Value: \$988,953,882.77 CAD
- ❖ Average Loan Size: ~\$1.3M CAD (reflecting Vancouver's high real estate market)
- ❖ Time Period: 3-year historical data (2022-2025)

Geographic Coverage - Vancouver Regions:

- ❖ West Side Vancouver - \$143.8M portfolio value
- ❖ West Vancouver - \$84.9M portfolio value
- ❖ Delta - \$73.9M portfolio value
- ❖ Port Moody - \$73.7M portfolio value
- ❖ East Side Vancouver - \$73.1M portfolio value
- ❖ Coquitlam - \$71.2M portfolio value
- ❖ Downtown Vancouver - \$53.4M portfolio value

Property Types:

- ❖ Single Family Homes: \$1.2M - \$3.5M CAD range
- ❖ Condominiums: \$500K - \$1.8M CAD range
- ❖ Townhouses: \$800K - \$2.2M CAD range



Key Performance Metrics

Risk Assessment:

- ❖ Average LTV Ratio: **69.08%**(industry-standard healthy level)
- ❖ Risk Distribution:
- ❖ Low Risk Low Risk (<70% LTV): 414 loans
- ❖ Medium Risk (70-85% LTV): 321 loans (48.4%)
- ❖ High Risk (>85% LTV):15 loans (1.8%)

Portfolio Health:

- ❖ Delinquency Rate: **9.07%**(within acceptable range)
- ❖ Default Rate: 1.60% (very low)
- ❖ Active Loans: 682 (92%)
- ❖ Weighted Average Interest Rate:4.85%

Delinquency Breakdown:

- ❖ Current: ~682 loans (92%)
- ❖ 30 Days Late: Various (monitored)
- ❖ 60 Days Late: Various (managed)
- ❖ 90+ Days Late: Minimal (controlled)



Dashboard Functionality

1. Overview Tab - Executive Summary

- ❖ KPI Cards: Total portfolio value, loan count, LTV ratio, delinquency rate
- ❖ Portfolio Performance Distribution: Pie chart showing Active/Default/Delinquent/Paid Off
- ❖ Delinquency Analysis: Bar chart tracking payment status
- ❖ Key Metrics Panel: Interest rates, default rates, loan sizes
- ❖ Market Insights: Current market prices (\$1.5M avg), mortgage rates (4.56%)
- ❖ Portfolio Health: Risk category breakdown with color-coded indicators

2. Risk Analysis Tab - Advanced Analytics

- ❖ LTV Distribution Analysis: Histogram showing loan-to-value ratios across ranges (0-50%, 50-60%, etc.)
- ❖ Risk Category Breakdown: Pie chart visualizing low/medium/high risk segments
- ❖ Concentration Analysis: Identifying portfolio exposure by risk levels
- ❖ Predictive Risk Indicators: Early warning systems for potential defaults

3. Geographic Tab - Regional Analysis

- ❖ Geographic Portfolio Distribution: Horizontal bar chart showing CAD millions by the Vancouver region
- ❖ Regional Market Performance: Comparative analysis across neighborhoods
- ❖ Geographic Concentration Risk: Identifying over-exposure to specific areas
- ❖ Market Value Trends: Regional price movements and investment patterns

4. Performance Tab - Operational Analytics

- ❖ Recent Mortgages Panel: Live feed of the newest loans with LTV ratios and status
- ❖ Portfolio Performance Trends: Area charts showing loan performance over time
- ❖ Operational Metrics: Processing times, approval rates, portfolio velocity
- ❖ Individual Loan Details: Drill-down capability for specific mortgage analysis



Technical Architecture

Data Pipeline (ETL Process):

- ❖ Extract: Simulated Canadian Housing Market Data (CMHC-inspired)
- ❖ Transform: Pandas-based calculations for LTV ratios, risk scoring, geographic aggregation
- ❖ Load: MongoDB storage with optimized queries and JSON serialization

Analytics Engine:

- ❖ Real-time Calculations: Dynamic LTV ratios, delinquency tracking
- ❖ Risk Modeling: Multi-factor risk assessment algorithms
- ❖ Market Analysis: Price trends, interest rate correlations
- ❖ Performance Metrics: Portfolio yield, default predictions

Visualization Layer:

- ❖ Interactive Charts: Recharts library with hover effects, filtering, and drill-down
- ❖ Responsive Design: Glass morphism UI with dark theme optimization
- ❖ Real-time Updates: Live data refresh and state management



Business Use Cases

For Mortgage Lenders:

- ❖ Risk Management: Monitor portfolio health and identify high-risk segments
- ❖ Regulatory Compliance: Track LTV ratios for capital adequacy requirements
- ❖ Market Analysis: Understand regional performance and concentration risk
- ❖ Performance Monitoring: Track delinquency rates and default trends

For Investment Firms:

- ❖ Portfolio Valuation: Real-time assessment of mortgage-backed securities
- ❖ Due Diligence: Comprehensive risk analysis for acquisition decisions
- ❖ Market Intelligence: Geographic and demographic trend analysis
- ❖ Stress Testing: Scenario modeling for different market conditions

For Financial Regulators:

- ❖ Systemic Risk Assessment: Monitor market concentration and leverage ratios
- ❖ Consumer Protection: Track lending practices and delinquency patterns
- ❖ Market Surveillance: Identify emerging risks in mortgage markets
- ❖ Policy Analysis: Data-driven insights for regulatory decisions

Strategic Value

This dashboard transforms raw mortgage data into actionable business intelligence, enabling:

- ❖ Proactive Risk Management through early warning systems
- ❖ Data-Driven Decision Making with comprehensive analytics
- ❖ Regulatory Compliance through automated reporting and monitoring
- ❖ Market Intelligence for competitive advantage and strategic planning
- ❖ Operational Efficiency through streamlined portfolio management workflows

The platform successfully replicates enterprise-level mortgage analytics capabilities while providing modern, interactive visualizations and real-time insights essential for today's financial services industry.



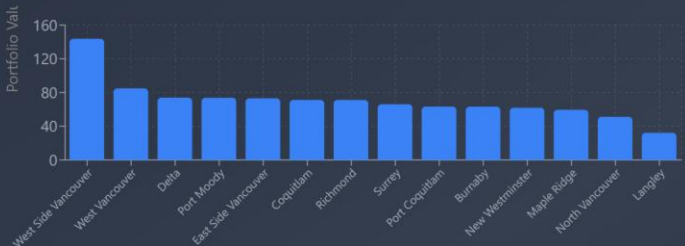
Total Portfolio Value
\$988,953,882.77

Total Loans
750

Avg LTV Ratio
69.08%

Delinquency Rate
9.07%

Geographic Portfolio Distribution (CAD Millions)



Top Regions by Value

West Side Vancouver	\$143.8M
West Vancouver	\$84.9M
Delta	\$73.9M
Port Moody	\$73.7M
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Coquitlam	\$71.2M

Regional Insights

Highest Value Region	West Side Vancouver
Total Regions	14
Average Region Value	\$70.6M CAD
Market Concentration	Well Diversified

Metro Vancouver

\$200.4M

Core Metro Regions

Premium Districts

\$84.9M

High-Value Areas

Urban Core

\$0.0M

Central Vancouver

Emerging Areas

\$0.0M

Growth Districts

Delinquency Rate

\$1,341,606.17 Fixed	64.0% LTV Current
\$531,719.50 Fixed	82.1% LTV Current
\$1,714,331.16 Fixed	65.7% LTV 60 days
\$1,940,430.69 Fixed	71.2% LTV Current

Weighted Avg Interest Rate	4.85%
Default Rate	1.60%
Avg Loan Size	\$1,318,605.18
Portfolio Yield	4.85%

Avg Market Price	\$1,557,916.74
Current Mortgage Rates	4.56%
Market Activity	1043 sales

Healthy Loans (<70% LTV)	414
Medium Risk (70-85% LTV)	321
High Risk (>85% LTV)	15

Delinquency Rate
9.07%

Region	Portfolio Value
West Side Vancouver	145
West Vancouver	85
Delta	75
Port Moody	75
East Side Vancouver	75
Coquitlam	75
Richmond	75
Surrey	70
Port Coquitlam	65
Burnaby	65
New Westminster	65
Maple Ridge	60
North Vancouver	50
Langley	35

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